

Descriptive Statistics

Data Analysis

2026

Preliminary Mathematics Standard

The Subject Matter of Statistics

Descriptive Statistics

Visual Summaries

Some Common Graphs and Charts

Quantitative Summaries

Homework

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The Subject Matter of Statistics

In statistics, **data**, are represented by random variables.

Definition

A **random variable** is a function, the value of which is a real number determined by the probability associated with the occurrence of each of its outcomes.

- random variables are **numeric** or **categorical**
- random variables are:
 - continuous** if its range forms an infinite set of real numbers i.e. consists of intervals *e.g. height or weight*
 - discrete** if its range is countable i.e. consists of individual (discrete) values that can be listed *e.g. # of rooms, shoe size*

*anything that can only be classified
e.g. colour of your hair*

*nominal: cannot be ordered
e.g. hair colour*

*ordinal: anything that can be ordered
e.g. shoe size, house number*

Discrete or Continuous?

- ① # of siblings ← Discrete
- ② Temperature ← Continuous
- ③ Body measurements ← Cont.
- ④ Hrs Slept ← Cont.
- ⑤ # of devious in a house ← Discrete

Normal or Ordinal?

- ① Place of Birth ← Nominal
- ② Date of Birth ← Ordinal
- ③ Income Bracket ← Ordinal
- ④ Fan Speed ← Ordinal
- ⑤ Car model ← Nominal
- ⑥ Song Genre ← Nominal

Numeric or Categorical?

- ① Quality of Presentation
(1 to 5) ← Numeric
(poor to high) ← categorical
- ② Fan Speed
(1 to 3) ← Numeric
(low to high) ← categorical

The Subject Matter of Statistics

Statistics is concerned with techniques for

- 1 **Descriptive Statistics:** the collection, presentation and summarisation of data
- 2 **Inferential Statistics:** the analysis of data for the purpose of making decisions in the face of uncertainty

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Descriptive Statistics

Descriptive statistics provide simple summaries about the data.
These summaries may be:

- visual: using tables, graphs and charts to display the data, or,
- quantitative: using statistical measures for:

single variable →

- **univariate data**: measures of central tendency (mean, median, mode) and measures of spread (variance, standard deviation, range, interquartile range)

two variables →

- **bivariate data**: correlation, line of best fit

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Frequency Tables

Both numerical and categorical data can be organised into a frequency table, numerical data can also be organised into a cumulative frequency table.

Frequency Tables

Example

A class of twenty students were asked how many pets they owned.

The results were:

3 0 1 4 4 1 2 0 2 2
0 2 0 1 3 1 2 1 1 3

discrete
ordinal
numeric

(∞)

f

Create a frequency table based on this data.

cumulative freq.

x	tally	f	cf
0		4	4
1		6	10
2		5	15
3		3	18
4		2	20
Sum (Σ)		20	

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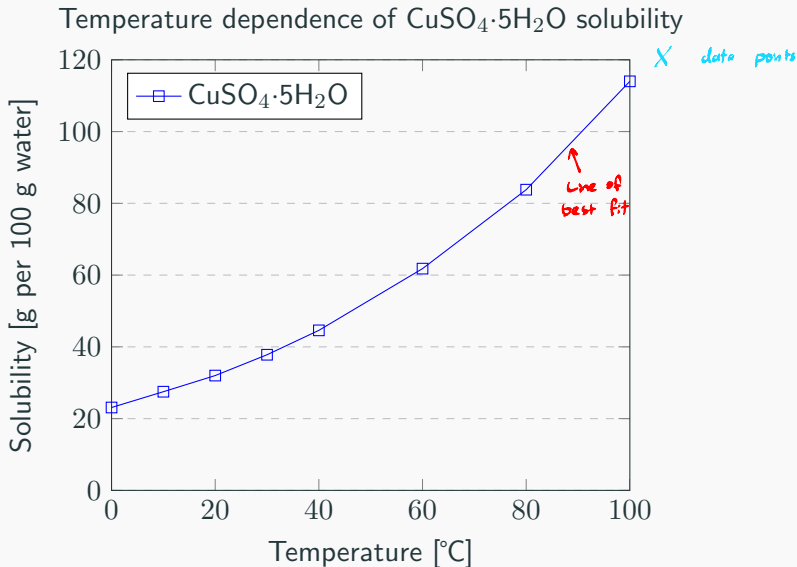
Visual Summaries

Some Common Graphs and Charts

Quantitative Summaries

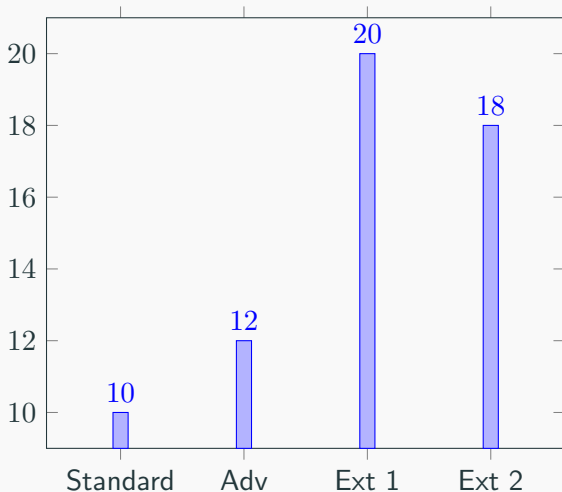
Homework

Line Graph

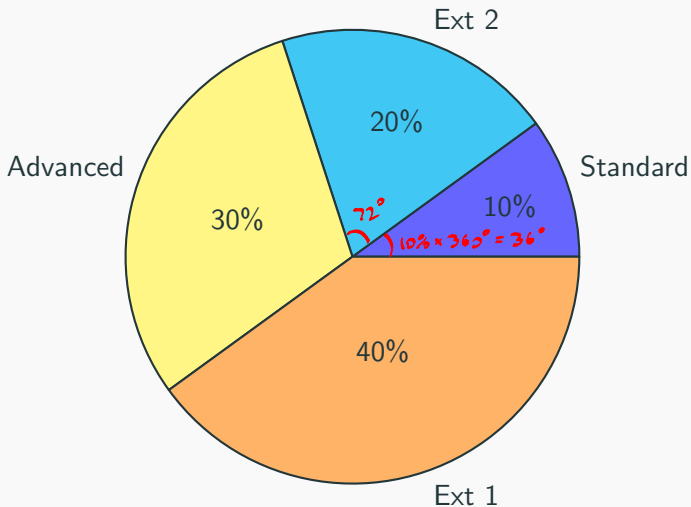


Column Graph

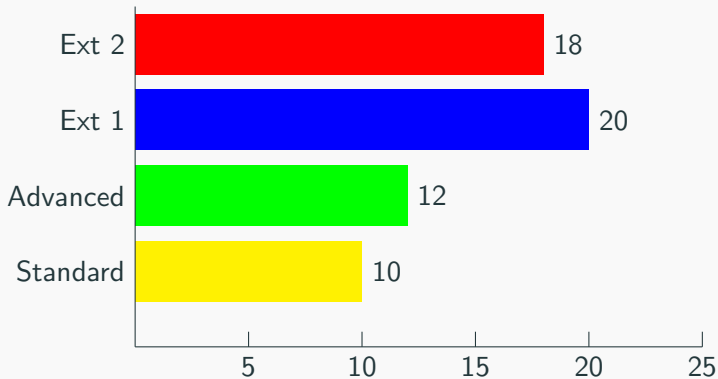
Number of Students by Mathematics Course



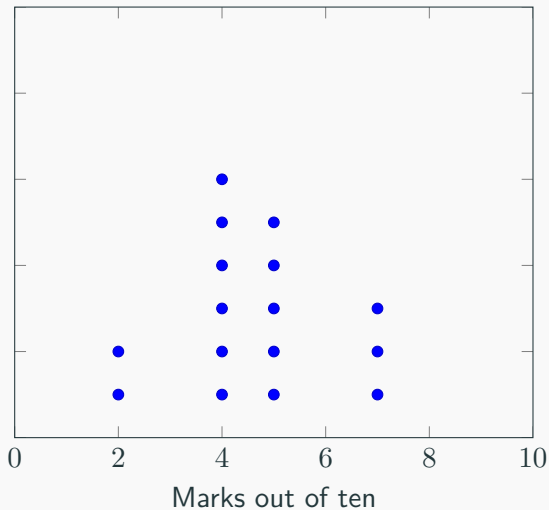
Pie Chart/Sector Graph



Bar Chart



Dot Plot



Pareto Chart

A Pareto Chart is a graph that indicates the frequency of defects, as well as their cumulative impact.

They are useful in finding the defects to prioritise in order to observe the greatest overall improvement.

It consists of two graphs drawn together

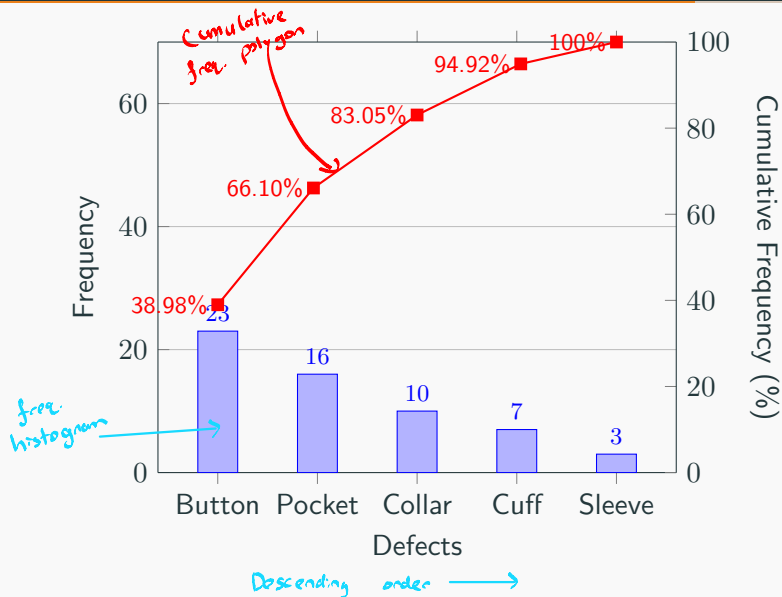
- a **frequency histogram** arranged in descending order
- a **cumulative frequency polygon** (ogive)

Pareto Chart

Example

Defect	Frequency	Cumulative Frequency	c.f (%)
Button	23	23	$23/59 = 39\%$
Pocket	16	39	$39/59 = 66\%$
Collar	10	49	$49/59 = 83\%$
Cuff	7	56	$56/59 = 94\%$
Sleeve	3	59	$59/59 = 100\%$
Σ	59		

Pareto Chart



Eliminating the button and pocket defects will remove 66% of the

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Two Measures of Central Tendency — Mode

The **mode** of a sample (dataset) is the score (datum) that occurs most often. In our example the mode is 1 pet.

The mode is not necessarily unique, the sample may be:

- bimodal (two scores with the greatest frequency)
- trimodal (three scores with the greatest frequency)
- multimodal (more than three scores with the greatest frequency)

Two Measures of Central Tendency — Median

Odd:

3 5 7 8 9 10 12
 ↑
 Median

Even:

3 3 5 7 8 9 10
 └─┬─┘
 7.5
 Median

The **median** of a sample is:

- For an odd number of scores: the middle score when the data is arranged in ascending order
- For an even number of scores: the average of the two middle scores when the data is arranged in ascending order.

In our example there are 20 scores, so the median is the average of the 10th and 11th scores i.e. 1.5 pets.

Mean: Arithmetic average

A Measure of Spread

The **range** of a sample is the difference between the maximum and minimum score.

Only numeric data will have a range. In our example the range is $4 - 0 = 4$ pets.

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Complete all questions from *Data Analysis — Descriptive Statistics*.